Transformations of Absolute-Value Graphs
Answer Key
Algebra 1

Quick Answers

1. (a) x-intercepts = -3, 3, (b) y-intercept = -3
2. (a) x-intercept = 4, (b) y-intercept = 4
3. (a) x-intercepts = -7, 3, (b) y-intercept = -3
4. (a) x-intercepts = -6, 6, (b) the value at x = 2 = 4
5. (a) x-intercepts = -4, 4, (b) the value at x = 3 = -2
6. (a) x-intercepts = -3, 5, —
7. (a) x-intercepts = -5, 3, (b) y-intercept = 6
8. (b) the value at the corner = 2, —
9. (a) x-intercepts = -5, 1, —
10. (a) x-intercepts = 0, 4, (b) the value at x = 5 = 3
11. (b) x-intercepts = -1, 5, —
12. (a) x-intercepts = -4, 8, (b) y-intercept = 2, —

What is verified

7 of 12 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 6, 8, 9, 11, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1
HSF-BF.B.3 – *problems 1, 2, 3, 4, 5, 7, 8, 10, 11*
HSF-IF.C – *problems 6, 9, 12*
Difficulty 1–5 of 5 across 25 tagged checks.

Common wrong answers

2. *If they got -4: read x - 4 as a shift of 4 to the left*

1. $f(x) = |x| - 3$. (a) x-intercepts. (b) y-intercept.

(a) Set $y = 0$: $|x| = 3$, and two numbers are three units from zero.

$$\boxed{x = -3 \quad \text{and} \quad x = 3}$$

(b) At $x = 0$: $|0| - 3$. The corner has been pulled straight down to $(0, -3)$. $\boxed{-3}$

2. $f(x) = |x - 4|$. (a) x-intercept. (b) y-intercept.

(a) $|x-4|=0$ only when the inside is zero, so there is a single crossing: the corner itself sits on the axis. The bracket reads "minus four" and the graph has moved four to the *right*. $\boxed{x=4}$

(b) At $x=0$: $|0-4|=|-4|$. $\boxed{4}$

3. $f(x)=|x+2|-5$. (a) x-intercepts. (b) y-intercept.

(a) $|x+2|=5$ splits into two cases: $x+2=5$ and $x+2=-5$.

$$\boxed{x=-7 \quad \text{and} \quad x=3}$$

Both are five units from the corner at $x=-2$, which is the check.

(b) At $x=0$: $|2|-5=2-5$. $\boxed{-3}$

4. $f(x)=-|x|+6$. (a) x-intercepts. (b) $f(2)$.

(a) $-|x|+6=0$ gives $|x|=6$. The minus in front turns the V upside down, so the graph comes *down* to meet the axis on both sides.

$$\boxed{x=-6 \quad \text{and} \quad x=6}$$

(b) $f(2)=-|2|+6=-2+6$. $\boxed{4}$

5. $f(x)=2|x|-8$. (a) x-intercepts. (b) $f(3)$.

(a) $2|x|=8$, so $|x|=4$. The 2 makes the arms twice as steep, so the graph reaches the axis sooner than $|x|-8$ would.

$$\boxed{x=-4 \quad \text{and} \quad x=4}$$

(b) $f(3)=2(3)-8=6-8$. $\boxed{-2}$

6. $f(x)=|x-1|-4$. (a) x-intercepts. (b) Sketch.

(a) $|x-1|=4$ gives $x-1=4$ or $x-1=-4$.

$$\boxed{x=-3 \quad \text{and} \quad x=5}$$

(b) — instructor-judged. The graph is a V opening upward with its corner at $(1,-4)$, read straight off the equation, and arms crossing the horizontal axis at -3 and 5. Both arms must be straight and equally steep — one up for one across, since the multiplier is 1. A curve, or a corner anywhere but $(1,-4)$, is not full credit.

7. $f(x)=-2|x+1|+8$. (a) x-intercepts. (b) y-intercept.

(a) $-2|x+1|=-8$, so $|x+1|=4$ and $x+1=\pm 4$.

$$\boxed{x=-5 \quad \text{and} \quad x=3}$$

(b) At $x=0$: $-2|1|+8=-2+8$. $\boxed{6}$

8. $f(x)=|x-3|+2$. (a) Describe the two moves from $y=|x|$. (b) The value at the corner.

(a) — instructor-judged. Two moves: three units to the *right* (the bars contain $x-3$, and the corner is where the inside is zero, at $x=3$), then two units up. Full credit needs both directions right. Reading $x - 3$ as a move to the left is the error this item exists to catch.

(b) The corner is at $x = 3$, and there $|3 - 3| + 2 = 0 + 2$. $\boxed{2}$

9. $f(x) = -|x + 2| + 3$. (a) x-intercepts. (b) Sketch.

(a) $|x + 2| = 3$, so $x + 2 = \pm 3$.

$$\boxed{x = -5 \quad \text{and} \quad x = 1}$$

(b) — instructor-judged. An upside-down V — the minus in front flips it — with the corner at $(-2, 3)$ and straight arms of equal steepness crossing the horizontal axis at -5 and 1. An upward V, or a corner at $(2, 3)$, is not full credit.

10. $f(x) = 3|x - 2| - 6$. (a) x-intercepts. (b) $f(5)$.

(a) $3|x - 2| = 6$, so divide by 3 *before* splitting: $|x - 2| = 2$, then $x - 2 = \pm 2$.

$$\boxed{x = 0 \quad \text{and} \quad x = 4}$$

Splitting before dividing is where this one goes wrong.

(b) $f(5) = 3|5 - 2| - 6 = 9 - 6$. $\boxed{3}$

11. A V has its corner at $(2, -3)$, opens upward, arms of slope ± 1. (a) Write the equation. (b) x-intercepts.

(a) — instructor-judged. The equation is $y = |x - 2| - 3$. Full credit needs both signs: a corner two units to the right puts $x - 2$ *inside* the bars, and a corner three units down subtracts 3 outside them. Writing $|x + 2|$ is the usual reversal and earns nothing here.

(b) $|x - 2| = 3$, so $x - 2 = \pm 3$.

$$\boxed{x = -1 \quad \text{and} \quad x = 5}$$

12. Synthesis challenge. $f(x) = -\frac{1}{2}|x - 2| + 3$. (a) x-intercepts. (b) y-intercept. (c) Sketch and explain the effect of the one-half.

(a) Undo the outside first: $-\frac{1}{2}|x - 2| = -3$, so $|x - 2| = 6$ and $x - 2 = \pm 6$.

$$\boxed{x = -4 \quad \text{and} \quad x = 8}$$

(b) At $x = 0$: $-\frac{1}{2}|-2| + 3 = -1 + 3$. $\boxed{2}$

(c) — instructor-judged. The graph is an upside-down V with its corner at $(2, 3)$ and arms reaching the horizontal axis at -4 and 8. The written part must say that the $\frac{1}{2}$ makes the arms *less* steep than the parent graph: the graph falls one unit for every two units sideways, so it takes six horizontal units to fall the three units from the corner to the axis. That is exactly why the crossings are six either side of 2 rather than three. A sketch with nothing said about steepness answers only half of (c).